

A multirotor drone is shown in flight, spraying a fine mist of liquid onto a field of rice crops. The drone is a quadcopter with a white body and black arms, equipped with a large white tank and multiple nozzles. It is positioned in the center-right of the frame. The background features a lush green landscape with palm trees and other tropical vegetation under a bright sky. The overall scene is a demonstration of agricultural drone technology.

SESSION S03

Multirotor Configuration and Theory

Robotics, Drones & Autonomous Systems

KAUST Gifted Student Program · Summer Enrichment 2026

A NEW PROJECT



The configuration class is given.
Its parameters are not.

Session outline

- How a multirotor works, and the physics that sizes it

Rotor count, thrust margin and redundancy — then momentum theory, disc loading and the scaling laws.

- New briefs. You choose the airframe.

Rotor count, layout and frame size, justified from your brief.

- Read-back

This session and S04 are **one project**. Here you choose the airframe; in S04 you choose what it is made of and find out whether it flies.

Output: Report 2 §1 — the airframe decision. The rest of Report 2 is S04's components and mass budget. **Together they are 20%.**

PART ONE



Principles of multirotor flight

Four fixed-pitch rotors, four axes of control

A multirotor cannot change the pitch of its blades. The only thing it can change is how fast each motor spins — and out of that it has to produce four independent controls.



Height

Speed all of them up together.



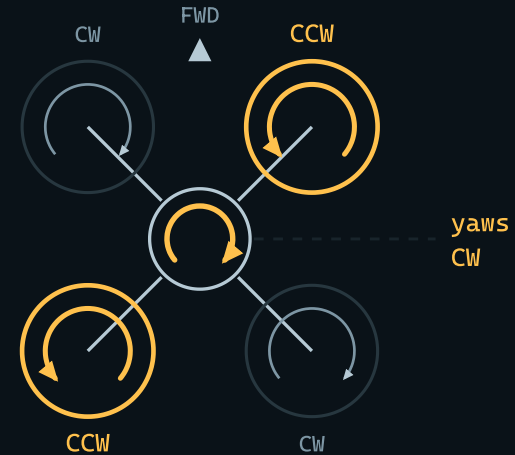
Roll and pitch

Speed up one side, slow the other. The aircraft tilts, and the tilted thrust pushes it sideways. A multirotor moves by leaning.



Yaw

Every spinning propeller drags the airframe the opposite way. Half spin one way, half the other, and it cancels — until you speed one *diagonal pair* up.



Speed up the **gold pair** and the frame turns the other way. Nothing tilts — that is yaw.

Hence even numbers with opposing spin directions — and a tricopter needing a tilting tail motor to get its yaw.

Rotor count

LAYOUT	ROTORs	LIFT FOR THE SIZE	SURVIVES A MOTOR LOSS	EFFICIENCY	COST
Tricopter	3	●●●●●	NO	●●●●●	●●●●●
Quadcopter	4	●●●●●	NO	●●●●●	●●●●●
Hexacopter	6	●●●●●	●●●●●	●●●●●	●●●●●
Octocopter	8	●●●●●	●●●●●	●●●●●	●●●●●
X8 · coaxial	8 on 4 arms	●●●●●	●●●●●	●●●●●	●●●●●

More rotors buy lift and redundancy. They cost efficiency, mass, cost and complexity — every motor needs its own ESC, its own wiring and its own arm.

The X8 is the interesting one: eight motors stacked in pairs on four arms. Same footprint as a quad, nearly the redundancy of an octo — and it pays for it, because a propeller working in the wake of another one is **roughly 15% less efficient**.

Thrust-to-weight ratio and control margin

THRUST MARGIN, AND WHAT A LEAN COSTS

$$\frac{T}{W} = \frac{n T_{\text{motor}}}{mg}$$

$$T_{\text{needed}} = \frac{mg}{\cos \varphi}$$

T/W Thrust-to-weight ratio, dimensionless
 T_{motor} Thrust one motor produces, N
 T_{needed} Thrust to hold height while leaning

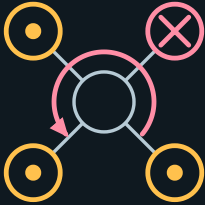
n Rotor count. Each motor makes $(T/W) \cdot mg/n$
 m, g Takeoff mass kg; gravity 9.81 m/s^2
 ϕ Lean angle. At 30° , 15% more thrust

- | | |
|---------|---|
| 1.5 : 1 | Barely flyable. No reserve for gusts, no authority to correct an upset. Not a design point |
| 2 : 1 | The normal target. Half your thrust holds you up, half is available to manoeuvre and to fight wind |
| 2.5 : 1 | Windy sites, precise position holding, or anything that must recover from an upset near a structure |
| 3 : 1 + | Aerobatic and racing. You are carrying motor mass you will never use on a survey |

Wind is paid for out of your thrust margin — a rotor tilted into a gust puts less of its thrust into holding you up.

Behaviour following a single motor failure

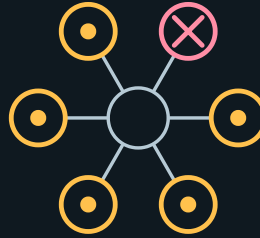
4 ROTORS



Does not survive

Three left, each +33% — but three knobs cannot answer four demands. Height, roll and pitch win; yaw is abandoned.

6 ROTORS



Survives, given sufficient margin

Five left, each +20%, and five knobs still cover four demands. The catch is thrust: it needs $T/W \geq 1.2$ to hold height on five.

8 ROTORS · X8



Survives, with margin to spare

Seven left, each +14%, needing only $T/W \geq 1.14$. This is the layout for an aircraft flying over what must not be hit.

"Redundant" is not a property of the rotor count alone. It is a property of the rotor count and the thrust margin together — and the margin is something you either budgeted for or did not.

▶ Hex flies on five 1:10

▶ Hex does not 0:17

PART TWO



The parameters that determine size

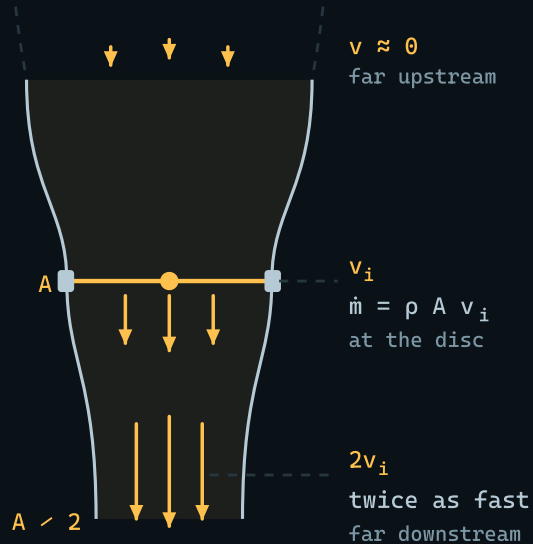
PAIRED DISCUSSION

What does it cost to hold an aircraft in the air?

One sentence, not a formula. The usual answer is "its weight" — but that is what you are holding, not what it costs to hold it.

60 SECONDS Twenty seconds alone, then compare with your neighbour

Hover power from momentum theory



Half the acceleration happens **above** the disc, half **below** it.

▶ See it in a wind tunnel 0:21

01 $\dot{m} = \rho A v_i$

Air through the disc each second

02 $T = \dot{m} \cdot 2v_i$

The wake gained $2v_i$, not v_i

03 $v_i = \sqrt{\frac{T}{2\rho A}}$

Big disc, gentle v_i

04 $P = T \cdot v_i$

Thrust $\times$ the speed you threw it at

IDEAL HOVER POWER · MOMENTUM THEORY

$$P_{\text{ideal}} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

and a real motor draws $P = P_{\text{ideal}} / (\text{FM} \cdot \eta)$

P Hover power, W

T Thrust, N. In hover $T = mg$

A Disc area, **all** rotors

v_i Induced velocity at the disc

$\dot{m}$ Air mass flow, kg/s

ρ , FM, η 1.225 · 0.65 · 0.85

Disc loading and its effect on endurance

DISC LOADING, AND WHAT IT COSTS YOU

$$DL = \frac{W}{A} \quad [\text{N/m}^2] \quad \Rightarrow \quad \frac{P}{W} = \frac{1}{FM} \sqrt{\frac{DL}{2\rho}}$$

DL Disc loading — weight carried per m² of disc

W Aircraft weight, **N** (= mg)

ρ Air density, **1.225 kg/m³**

P/W Power per newton of aircraft. **Lower is better**

A Total disc area, **m²**

FM Figure of merit, **≈0.65**

THE THEORY

Power per unit weight rises with $\sqrt{DL}$.

Halve the disc loading and you save about 30% of your hover power — at exactly the same takeoff weight. Nothing else you can do is worth that much.

→

APPLIED · A 3 KG AIRCRAFT

Four 10" props: $A = 0.20 \text{ m}^2$, $DL = 145 \text{ N/m}^2 \rightarrow$ about 410 W

Four 15" props: $A = 0.46 \text{ m}^2$, $DL = 65 \text{ N/m}^2 \rightarrow$ about 273 W

Same aircraft, same mass. A third less power.

This is S01's $\sqrt{A}$ with a name and a number attached — and it is why a survey multirotor looks nothing like a racing one.

The propeller scaling laws

THRUST AND POWER AGAINST DIAMETER AND SPEED

$$T = C_T \rho n^2 D^4$$

$$P = C_P \rho n^3 D^5$$

T Thrust from one propeller, N

n Revolutions per second — not per minute

ρ Air density, 1.225 kg/m³

P Shaft power it absorbs, W

D Propeller diameter, m

C_T , C_P Thrust and power coefficients — see the figure

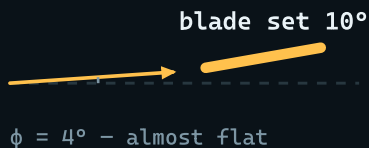
AT QUARTER SPAN

$$\Omega r = 20 \text{ m/s}$$



AT THE TIP

$$\Omega r = 80 \text{ m/s}$$



Dashed line is the **plane of rotation**. ϕ collapses as Ωr grows — hold the same α at both stations and the blade angle must fall with it. That is the **11° of twist**.

Go bigger, or go faster — diameter wins whenever you have room.

Where C_T comes from. Each blade is a wing — S01's airfoil, going round at local airspeed $\Omega \cdot r$. Four times the speed at the tip would be **sixteen times the lift** at the same angle; the twist is what stops that.

Tip speed is $\pi \cdot D \cdot n$. Past 200 m/s the tips go transonic. A big propeller must turn slowly.

PART THREE



Exercise — 35 minutes

Three new problem statements

BRIEF D · ROOFTOP LIFT

Four kilograms of plant equipment, 60 m up, twelve times a day — on a route that **crosses a public street**.

BRIEF E · SOLAR SURVEY

Thirty hectares of photovoltaic array, thermal and visual, **in one morning on three battery sets**.

BRIEF F · BRIDGE AND CULVERT

Fourteen structures, undersides and interiors, **inside a culvert 2.5 m wide** — where contact with concrete is expected, not feared.

Every one of these is a multirotor. **Which multirotor is the question** — and each brief has a different parameter doing the deciding.

LAYOUT X Arms at the corners. Nothing ahead of the camera, every motor works roll *and* pitch. **The default.**

LAYOUT + One arm forward, so a propeller sits in the shot. Rare on anything carrying a camera.

LAYOUT H Twin booms, a clear bay down the centre. Heavier for the same span. Used when the payload is *long*.

Task

CORE · EVERYONE FINISHES THIS

STRETCH · IF YOU FINISH EARLY

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- What would have to change for you to add two more rotors? Or remove two?

Review — two and a half minutes per team

Tell the room:

- 1 The one parameter that decided your airframe
- 2 Your rotor count and layout, and what you rejected
- 3 Your thrust-to-weight target, and where it came from
- 4 What happens to your aircraft when one motor stops

Three teams, three briefs, and there is **no reason at all** for you to arrive at the same airframe. If two of you did, one of you has copied a default rather than read a brief.

NEXT SESSION

S04 — Component Selection

You have chosen an airframe. In S04 you find out whether real motors, real propellers and a real battery can lift it — for as long as your brief demands.

Your rotor count becomes the **thrust you must find**.

Your propeller size becomes the **power you must carry**.

Bring this sheet. Some of you are going to have to change it.

Photographs

Rice crop spraying — Christopher Hedreyd, PIA CALABARZON, public domain

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